

T-030 / [NEST-DEPTH]: a covering-number (few-circles) sufficient

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1. Purpose and scope

The in-bundle reproof attempt dr2-t030-bd-discrete-reproof v1.1 (R-036) split the discrete decoupling residual as

$$[\text{BD-DISCRETE}] = [\text{BD-CONTINUOUS}] + [\text{NEST-DEPTH}],$$

closing the poly-separated subclass ($s \geq N^{-C}$) modulo the textbook continuous theorem and naming *[NEST-DEPTH]* – control of the nested-cluster recursion depth – as “the single genuine open piece”. Its witness is the lacunary configuration $\{2^{-j} : j = 0, \dots, J\}$ carried onto a great circle, with minimal separation $s = 2^{-J}$ and recursion depth $J = \log_2(1/s)$, unbounded at fixed N ; iterating the one-step decoupling inequality at a fixed scale across J nested generations costs $D_0^{2^J} = \exp(\Theta(J))$.

This note asks the prior question the reproof note did not settle: **does the witness witness hardness of the BOUND, or only of the TECHNIQUE?** We show the latter, via an elementary covering-number bound, and use it to delineate the genuine open core. **Not claimed:** any unconditional arbitrary- Q closure; any tier/gate flip. The track is the H-NONLATTICE-REMAINDER frontier and is NOT load-bearing for the published C_{full} head (synthesis Sec. 4b).

2. The per-pair and single-circle bounds (R-021, T7 – prior)

We recall, with explicit attribution, the input on which everything below rests.

Lemma 1 (R-021, sphere-circle cross-energy, T7). *For finite A, B each supported on a circle of S^2 (planar section), the rectangular additive energy satisfies $E_+(A, B) := \sum_m r_{A,B}(m)^2 \leq 3|A||B|$, where $r_{A,B}(m) = \#\{(a, b) \in A \times B : a + b = m\}$. It holds for parallel and non-parallel circles.*

Mechanism (R-021). For fixed m , $a + b = m$ with $a \in \Gamma_A$ forces $a \in \Gamma_A \cap (m - \Gamma_B)$; the reflected-translated set $m - \Gamma_B$ is a circle, and two *distinct* circles in $\mathbb{R}^3$ meet in ≤ 2 points, so $r_{A,B}(m) \leq 2$ except for the at most one m_0 with $m - \Gamma_B = \Gamma_A$ (which forces equal radius and parallel planes, fixing m_0), where $r_{A,B}(m_0) \leq \min(|A|, |B|)$. Hence $E_+(A, B) \leq \min(|A|, |B|)^2 + 2|A||B| \leq 3|A||B|$.

Corollary 1 (single circle, T7 – prior). *If Q is supported on ONE circle then $E_+(Q) = E_+(Q, Q) \leq 3N^2$.*

This already disposes of the rich-latitude obstruction of R-025 (a latitude with $T' = N$): one rich circle, however rich, has $E_+ \leq 3N^2$. The aligned term is the center-reflection sum $m_0 = 2c$ (c = circle centre): for the great circle $c = 0$, $m_0 = 0$ and $r(0) = N$ are the antipodal pairs, giving $E_+ \leq (2N^2 - N) + (N^2 - N) < 3N^2$. (Machine: a rich latitude $x^2 + y^2 = 1105^2$ at $z = 47$, $N = 108$, has $E_+ = 34668$, $E_+/N^2 = 2.972$, exactly the published R-033 values F1/F5; assert A1.)

3. Theorem: the covering-number (few-circles) bound (new assembly)

Theorem 1 (few-circles additive-energy bound). *Let Q be a finite set covered by L circles, $Q \subset \bigcup_{i=1}^L \Gamma_i$. Then*

$$E_+(Q) \leq 3L^2 N^2, \quad N = |Q|.$$

Proof. Fix a partition $Q = \bigsqcup_{i=1}^L Q_i$ with $Q_i \subset \Gamma_i$ (assign each point to one covering circle). For every m , $r(m) = \#\{(a, b) \in Q^2 : a + b = m\} = \sum_{i,j=1}^L r_{ij}(m)$ with $r_{ij}(m) = \#\{(a, b) \in Q_i \times Q_j : a + b = m\}$, the index pair being unique per ordered pair (a, b) . Cauchy–Schwarz over the L^2 index pairs gives $r(m)^2 \leq L^2 \sum_{i,j} r_{ij}(m)^2$, and summing over m ,

$$\begin{aligned} E_+(Q) &= \sum_m r(m)^2 \leq L^2 \sum_{i,j} \sum_m r_{ij}(m)^2 = L^2 \sum_{i,j} E_+(Q_i, Q_j) \\ &\stackrel{\text{Lemma R-021}}{\leq} 3L^2 \sum_{i,j} |Q_i| |Q_j| = 3L^2 \left(\sum_i |Q_i| \right)^2 = 3L^2 N^2. \quad \square \end{aligned}$$

Corollary 2 (a new unconditionally-closed class). *If $Q \subset S^2$ is coverable by $L \lesssim_\epsilon N^\epsilon$ circles then $E_+(Q) \lesssim_\epsilon N^{2+\epsilon}$ – DR-2 holds UNCONDITIONALLY for Q , with no decoupling theorem and no circle-incidence bound.*

The bound is elementary (R-021 + a Cauchy–Schwarz packaging) and is informative only for $L = N^{o(1)}$: at $L \sim N^{1/2}$ it degrades to $3N^3$. Its role is to add a covering-number axis to the existing sufficient conditions, complementary to the richness axis of Lemma A. Machine: the bound holds on every tested configuration (asserts A2, A3, A8), including a maximally-aligned same-radius $L = 4$ stack ($E_+ = 1,525,392 = E_+(\text{circle}) \cdot E_+(\text{AP}_4) = 34668 \cdot 44$, ratio $E_+/N^2 = 8.17 \leq 48 = 3 \cdot 4^2$) and an $L = 3$ distinct-radius union (ratio $2.45 \leq 27$).

4. The [NEST-DEPTH] witness is closed at $L = 1$ (technique-only)

Proposition 1 (witness benignity). *The [NEST-DEPTH] lacunary witness $\{2^{-j}\}_{j=0}^J$ carried onto a great circle is supported on ONE circle. By the Corollary of Sec. 2, $E_+ \leq 3N^2$ for every J , independent of the recursion depth $J = \log_2(1/s)$.*

Consequently the $D_0^{2J} = \exp(\Theta(J))$ blow-up that the reproof note exhibited is a property of the *fixed-scale decoupling iteration*, not of the additive energy: the witness that motivated [NEST-DEPTH] does not witness any hardness of the target bound – it has minimal $(\Theta(N^2))$ energy. Machine (assert A6): with rational points $t_j = 2^{-j}$ on $x^2 + y^2 = 1$, exact integer energy gives

J	N	depth = $\log_2(1/s)$	E_+/N^2
4	5	3	1.800
5	6	4	1.833
6	7	5	1.857
7	8	6	1.875

the depth grows while E_+/N^2 stays flat near 2 (< 3). **Reclassification.** The genuine open content of [NEST-DEPTH] is NOT the lacunary / near-one-circle family (benign by R-021) but the truly two-dimensional clustered family that requires *many* circles to cover – exactly the regime Sec. 5 isolates.

5. Sharp delineation of the open core

Collecting the unconditional sufficient conditions for arbitrary finite $Q \subset S^2$:

$$E_+(Q) \leq \min \left\{ 3L^2 N^2 \text{ (this note)}, (1 + T') N^2 \text{ (R-025)}, C N^{9/4} \text{ (R-033, mod [CIRC-INC])} \right\},$$

with the lattice class (R-026, $T' \lesssim_\epsilon R^\epsilon$) and the poly-separated class (R-037, modulo [BD-CONTINUOUS]) closed separately. A family can therefore violate $E_+ \lesssim_\epsilon N^{2+\epsilon}$ only if it is SIMULTANEOUSLY (a) covered by no $N^{o(1)}$ circles ($L = N^{\Omega(1)}$), (b) rich, $T' = N^{\Omega(1)}$, and (c) beyond the reach of the $N^{9/4}$ incidence count. By R-033 Cor. 1.2 such a family must carry $\geq N^{2+\delta}/(8k^2 \log N)$ simultaneously k -rich sum-circles – a super-polynomial number of distinct rich circles. The few-circles bound shows this multiplicity is *necessary*: a bounded (or polynomial) circle cover is incompatible with $E_+ \geq N^{2+\delta}$. The open core of T-030 is thus precisely the **super-polynomially-many-simultaneously-rich-circles regime**; the lacunary/few-circle families that previously appeared as obstructions are not in it.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

- (α) “*This is R-021 plus a Cauchy–Schwarz; the per-pair and single-circle bounds are not new.*” VALID, fully acknowledged in Sec. 2: R-021 (T7) gives $E_+(A, B) \leq 3|A||B|$ and hence the $L = 1$ single-circle bound. The new content is exactly (i) the covering-number assembly $3L^2N^2$ (R-021 is pairwise only; the L^2 Cauchy–Schwarz over index pairs is the packaging), (ii) the resulting covering-number sufficient class, (iii) the reclassification of the [NEST-DEPTH] witness as technique-only via $L = 1$, and (iv) the open-core delineation. No deep new analysis is claimed; the deliverable is a structural sharpening plus one new closed sub-class.
- (β) “*The L^2 factor is lossy – at $L \sim N^{1/2}$ the bound is $3N^3$, vacuous.*” VALID and central to the honest scope: the theorem is effective ONLY for $L = N^{o(1)}$ and explicitly does NOT touch the many-circle core, which is named as the residual (Sec. 5). No closure of T-030 is claimed.
- (γ) “*Every Q is covered by some circles, so the hypothesis loses no generality / the class is trivial.*” DISMISSED with scope: every Q is covered by $L \leq N$ circles trivially, but the bound is non-vacuous only for $L = N^{o(1)}$; it is a SUFFICIENT condition delimiting a genuine sub-class (latitude unions, bounded-degree-curve sets, the witness), and its complement – the super-poly-circle regime – is carried as the open core, not silently included.
- (δ) “*The witness-benignity secretly uses the great circle’s $m = 0$ antipodal structure.*” DISMISSED: $E_+(Q) \leq 3N^2$ for Q on ANY single circle is R-021’s diagonal, valid for great AND non-great circles; the conclusion needs only “one circle”. Verified on a rich non-great latitude ($z = 47$, $E_+/N^2 = 2.972 \leq 3$; assert A1) as well as the great circle ($z = 0$).

Quantitative sanity check (mandatory). *Magnitude / constant.* The single rich circle has measured $E_+/N^2 = 2.972$ ($N = 108$, $E_+ = 34668$, matching the independently-published R-033 F1/F5 values exactly) – inside and near the proven constant 3, so the constant is correct, not loose by orders. *Exact limit case.* The $L = 4$ same-radius stack factorises exactly, $E_+ = 34668 \cdot 44 = 1,525,392 \leq 8,957,952 = 3 \cdot 4^2 N^2$ (margin $5.9\times$). *Depth direction.* The witness ratio is flat ($1.800 \rightarrow 1.875$) as the depth grows $3 \rightarrow 6$ – the predicted depth-independence. *Estimator audit.* The vectorised E_+ equals the brute $O(N^4)$ count ($396 = 396$) on a 12-point circle. *Reproducibility.* All counts are exact integer; the driver is deterministic and exits 0 iff all 8 asserts pass.

7. Result footer

Result ID: R-038 (few-circles / covering-number additive-energy bound + [NEST-DEPTH] witness reclassification) / T-030 frontier

Precise statement: (1) Q covered by L circles $\Rightarrow E_+(Q) \leq 3 L^2 N^2$ (Cauchy–Schwarz over L^2 index pairs + R-021 per-pair

$E_+(Q_i, Q_j) \leq 3|Q_i||Q_j|$. Cor: $L \leq \epsilon N^\epsilon \Rightarrow E_+ \leq \epsilon N^{2+\epsilon}$, arbitrary-Q DR-2 closed UNCONDITIONALLY for the few-circles class.
 (2) The [NEST-DEPTH] lacunary witness (great circle, $L=1$) has $E_+ \leq 3N^2$ INDEPENDENT of nesting depth $\log_2(1/s)$: the $\exp(\text{depth})$ telescoping loss is a fixed-scale-technique artifact, not bound hardness. Witness reclassified TECHNIQUE-ONLY.
 (3) Open core delineated: T-030 open only for Q with circle-cover $L = N^{\{\Omega(1)\}}$ AND $T' = N^{\{\Omega(1)\}}$ (super-poly-many simultaneously-rich circles; R-033 Cor.1.2).
 Scope: T-030 (arbitrary finite Q in S^2 ; B5 hypothesis H-NONLATTICE-REMAINDER-EXCLUDED). NOT load-bearing for the published C_{full} head (Lemma 2 caps $T' \leq 10$ in-class; synthesis Sec.4b).
 Evidence grade: Few-circles bound: T7 WITHIN-CLASS (elementary; R-021 [T7] + Cauchy-Schwarz). Witness reclassification: PROVED (R-021 at $L=1$ + machine depth-independence). Open-core delineation: structural synthesis. Does NOT close T-030: $E_+ \leq \epsilon N^{2+\epsilon}$ for arbitrary Q STILL OPEN (unconditional record $N^{\{9/4\}}$, R-033).
 Reproduction command: `python3 codes/vacuum/dr2_t030_fewcircles.py`
 Expected output: 8/8 asserts PASS, exit 0; JSON artefact at `claims/B5-BEYOND-LAYER-BOUND/runs/260614-dr2-t030-fewcircles/result.json`
 Falsifier: (a) Q on L circles with $E_+ > 3L^2N^2$ (refutes the theorem / R-021); (b) a single-circle family with E_+/N^2 unbounded (refutes R-021); (c) a lacunary witness with E_+ growing with depth at fixed N (refutes benignity). All three machine-checked false.
 Tier before / after: NONE -- operator decides. No gate/tier/hypothesis flip. B5 stays T7-SCOPE_{admissibility-bounded} given A1. T-030 stays OPEN; DR2-SHARE stays RESCOPED-T0-T030-NONLOADBEARING.
 No-overclaim statement: This note does NOT close T-030. The per-pair and single-circle bounds are R-021 (prior, T7), credited explicitly; the new content is the covering-number assembly, the witness reclassification, and the open-core delineation. The L^2 factor makes the bound vacuous for $L = N^{\{\Omega(1)\}}$ -- exactly the open core.
 Next required action: Operator review. Research continuation options:
 (i) bound the circle-cover number L by structural invariants of high-energy Q (would shrink the open core); (ii) attack the super-poly-rich-circle core via improved circle incidences (R-033 route, below 20/9); (iii) record this as the settled status of the [NEST-DEPTH] witness (technique-only) in the DR-2 programme map.